Lecture 21 22 Revision

L21_L22 (2).pdf
PDF

Prepare a clear, well-structured, and exam-focused set of lecture notes for lectures 21 and 22. The notes should be suitable for direct revision in a closed-book or reading-based exam.
You can use only the content of the Lecture 21 and 22 slides/PDF, and the text from the attached PDF. Do not include external explanations, additional explanations, or new examples beyond the given content. Clearly present all key definitions, notations, notions, properties, theorems, standard results, and important formulas covered in the tutorial. Maintain a formal academic tone and appropriate mathematical notation throughout. Cover all relevant concepts without omitting any material from the lecture, and avoid oversimplification. The final result should be concise but comprehensive, specially designed for effective exam revision.

Below are **exam-focused revision notes** for **Lectures 21 and 22** based **only** on the uploaded lecture material/PDF. I've kept them **formal, compact, and complete for revision**, while preserving the lecture structure and examples from the slides. Source:

CSE400 – Lecture 21 & 22 Revision Notes
Inequalities and Tail Bounds

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1. Tail of a Random Variable

Definition

The **tail** of a random variable X is:

$$P(X > x)$$

It measures the probability that X takes a **large / extreme value**.

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2. Why Do We Care About Tails?

Motivation

Tail probabilities are useful for studying **rare but important events**, for example:

- Jobs delayed more than 24 hours
- Packets dropped due to a full buffer
- A server receiving unusually high load

Key Idea

- **Mean** $\longrightarrow$ average behavior (**easy** to compute)
- **Tail** $\longrightarrow$ rare extreme events (**hard** to compute)

Why Are Tails Hard to Compute?

Because they often require summing **many small probabilities**.

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3. Tail Examples

Example 1: Random Job Assignment

Suppose n jobs are distributed among n servers at random.

Question

What is the probability that a particular server gets at least k jobs?

Model

$$X \sim \text{Binomial}(n, p)$$

Then:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Interpretation

This is a **tail event**: one server gets **many jobs** while most get few.

Example 2: Poisson Arrivals

Jobs arrive at a datacenter according to a **Poisson process** with rate λ jobs/hour.

Question

What is the probability that at least k jobs arrive during the first hour?

Model

$$X \sim \text{Poisson}(\lambda)$$

Then:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Interpretation

This is a **tail event** corresponding to a **burst of arrivals**.

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4. Why Tail Bounds?

Exact tail probabilities are often **difficult to compute** because they involve long sums such as:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

or

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Main Idea

Instead of computing the exact value, find an **upper bound**:

$$P(X \geq k) \leq \text{something simple}$$

Purpose of Tail Bounds

- Easier to compute
- Still provide a **guarantee**
- Useful for **safe approximation**

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5. Running Example Used Throughout

We repeatedly study the same experiment:

Experiment

Flip a fair coin n times.

Random Variable

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

Mean

$$E[X] = \mu = \frac{n}{2}$$

Question

Find / bound:

$$P\left(X \geq \frac{3}{4}n\right)$$

This is a **tail event**: getting **unusually many heads**.

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6. Markov's Inequality

6.1 Key Idea

Markov's inequality uses **only the mean**.

If the average is not too large, then **very large values must be rare**.

6.2 Statement

For a **non-negative** random variable $X \geq 0$ and any $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

This is a **right-tail upper bound**.

6.3 Proof Idea

Start from:

$$E[X] = \sum_{x=0}^{\infty} x\, p_X(x)$$

Restrict only to values $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x\, p_X(x)$$

Since $x \geq a$ on this region:

$$E[X] \geq \sum_{x=a}^{\infty} a\, p_X(x)$$

$$E[X] \geq a\, P(X \geq a)$$

Hence:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

6.4 Markov on Running Example

For

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right), \qquad \mu = \frac{n}{2}$$

we want:

$$P\left(X \geq \frac{3}{4}n\right)$$

Using Markov:

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{\mu}{\frac{3}{4}n}$$

$$= \frac{n/2}{3n/4} = \frac{2}{3}$$

Observation

This is a **poor bound**, because it **does not depend on** n.

6.5 Markov: Real-Life Example

Scenario: Income Distribution

Average income:

$$E[X] = 10{,}000$$

Question

What fraction of people earn at least $50{,}000$?

Using Markov:

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000} = 0.2$$

Result

At most **20%** of people can earn $50{,}000$ or more.

6.6 Markov: Solved Example 1

Problem (Exam Grades)

Average exam grade = 70%

"A" cutoff = 90%

Find an upper bound on the fraction of students who receive an "A".

Solution

Let $X \geq 0$ be the student grade.

$$\mu = E[X] = 70$$

By Markov:

$$P(X \geq 90) \leq \frac{E[X]}{90}$$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9} \approx 0.778$$

Result

At most **77.8%** of students can receive an "A".

6.7 Markov: Solved Example 2

Problem (Server Request Load)

A server processes an average of $5{,}000$ requests per second.

Find an upper bound on the probability that the server receives $20{,}000$ or more requests in a second.

Solution

Let $X =$ requests/sec.

$$E[X] = 5000, \qquad a = 20{,}000$$

By Markov:

$$P(X \geq 20{,}000) \leq \frac{5000}{20{,}000} = \frac{1}{4} = 0.25$$

Result

At most **25%** chance of $\geq 20{,}000$ requests.

6.8 Markov: Homework Problem

Problem

Average annual income is $40{,}000$.

(a) Bound:

$$P(X \geq 120{,}000)$$

(b) Can Markov bound:

$$P(X < 10{,}000)?$$

Hints from Lecture

- For (a), apply:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

with $a = 120{,}000$

- For (b), **Markov only bounds the right tail**, i.e. $P(X \geq a)$, not the left tail directly.

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7. Chebyshev's Inequality

7.1 Key Idea

Chebyshev's inequality uses:

- **Mean**
- **Variance**

It controls how far X can deviate from its mean.

7.2 Statement

For any random variable X with mean μ and variance $\mathrm{Var}(X)$, and any $a > 0$,

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Equivalent form using $\sigma^2 = \mathrm{Var}(X)$:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

This bounds the probability that X lies **far from the mean**.

7.3 Proof Idea

Apply Markov's inequality to the non-negative random variable:

$$(X - \mu)^2$$

Then:

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Using Markov:

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

Since:

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

we get:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

7.4 Chebyshev on Running Example

For

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right)$$

we have:

$$\mu = \frac{n}{2}, \qquad \mathrm{Var}(X) = \frac{n}{4}$$

We want:

$$P\left(X \geq \frac{3}{4}n\right)$$

Since the distribution is symmetric in the lecture treatment:

$$P\left(X \geq \frac{3}{4}n\right) = \frac{1}{2}P\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right)$$

Applying Chebyshev:

$$P\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right) \leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Hence:

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{1}{2} \cdot \frac{4}{n} = \frac{2}{n}$$

Key Insight

The bound **decreases as n increases**, so it is **better than Markov**.

7.5 Chebyshev: Real-Life Example

Scenario: Exam Scores

Mean score:

$$\mu = 70$$

Standard deviation:

$$\sigma = 10$$

Question

What fraction of students score below 40 or above 100?

This is:

$$P(|X - 70| \geq 30)$$

Using Chebyshev:

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2}$$
$$= \frac{100}{900} = \frac{1}{9}$$

Result

At most $\frac{1}{9}$ of students are this far from the mean.

7.6 Chebyshev: Solved Example 1

Problem (Exam Grades with Variance)

Average grade = 70%

Standard deviation = 5%

"A" cutoff = 90%

Find a tighter upper bound on the fraction of "A" students.

Solution

Given:

$$\mu = 70, \qquad \sigma = 5, \qquad \sigma^2 = 25$$

Observe:

$$X \geq 90 \implies |X - 70| \geq 20$$

Thus:

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Apply Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

So:

$$P(|X - 70| \geq 20) \leq \frac{25}{400}$$
$$= \frac{1}{16} = 0.0625$$

Result

$$P(X \geq 90) \leq 0.0625$$

This is **much tighter** than Markov's bound 0.778.

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8. Chernoff Bound

> The lecture outline explicitly includes **Chernoff Bound and Poisson Tail**, followed by definition, proof, and example.

From the lecture flow, Chernoff bounds are introduced as a **stronger tail bound** than Markov and Chebyshev for upper-tail probabilities, especially for sums of independent random variables and Poisson-type tails. The progression in the lecture is:

- Markov $\rightarrow$ mean only

- Chebyshev $\longrightarrow$ mean + variance
- Chernoff $\longrightarrow$ tighter exponential tail control

Conceptual Role in the Lecture

Chernoff bounds are used when one wants a **significantly tighter upper bound** on probabilities of large deviations such as:

$$P(X \geq k)$$

especially in settings like:

- many coin flips,
- Poisson arrivals,
- overload probabilities.

Exam Revision Takeaway

For revision, remember the **hierarchy of tail bounds** emphasized by the lecture:

$$\text{Markov} \ < \ \text{Chebyshev} \ < \ \text{Chernoff}$$

in terms of **tightness / usefulness for tails**.

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9. Poisson Tail

The lecture also explicitly covers **Poisson Tail**.

For a Poisson random variable:

$$X \sim \text{Poisson}(\lambda)$$

the upper tail is:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Interpretation

This gives the probability of seeing **many arrivals** in a fixed interval, such as:

- bursts in datacenter jobs,
- spikes in server demand,
- sudden queue buildup.

Exam Point

This exact expression is often **hard to compute directly**, which motivates using **tail bounds** instead.

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10. Jensen's Inequality

> The lecture outline explicitly includes **Jensen's Inequality**, **Convex Functions**, **Proof and Example**.

10.1 Convex Function

A function $g(x)$ is **convex** if it curves upward in the sense used in Jensen's inequality. The lecture presents this concept before Jensen's statement.

10.2 Jensen's Inequality (Lecture Topic)

Jensen's inequality connects:

- the expectation of a transformed random variable,
- and the transform of the expectation.

It is used in the lecture as a **general inequality tool** that supports reasoning about nonlinear functions of random variables.

Revision Point

For exam preparation, Jensen should be remembered as an inequality involving:

- **random variables**
- **expectation**
- **convex functions**

and it belongs to the broader family of **probabilistic inequalities** developed in these lectures.

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11. Case Study / System-Level Understanding

The lecture concludes with a **case-study / system-level understanding** section, including the question:

"How to predict the worst-case demands on server?"

System-Level Interpretation of Tail Bounds

Tail bounds are useful in systems because they help estimate the probability of **rare overload events**, such as:

- too many requests arriving,
- too many jobs landing on one server,
- queues growing unexpectedly,
- service delay or resource exhaustion.

Main Engineering Message

The lectures emphasize that **average-case behavior is not enough**.

In system design, we must also understand the **tail**:

$$P(X \geq \text{large threshold})$$

because these are the events that often cause:

- failures,
- congestion,
- overload,
- poor quality of service.

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12. Final Exam-Focused Summary

12.1 Core Definitions

- **Tail of X**:

$$P(X > x)$$

- **Right tail**:

$$P(X \geq a)$$

- **Deviation from mean**:

$$P(|X - \mu| \geq a)$$

12.2 Key Exact Tail Forms

Binomial Tail

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson Tail

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

12.3 Markov's Inequality

Condition

$X \geq 0$

Formula

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Uses

- Mean only
- Simple
- Often loose

12.4 Chebyshev's Inequality

Formula

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Uses

- Uses mean + variance
- Bounds deviations from mean

- Better than Markov in many cases

12.5 Running Example

For

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right), \qquad E[X] = \frac{n}{2}, \qquad \text{Var}(X) = \frac{n}{4}$$

Question

$$P\left(X \geq \frac{3}{4}n\right)$$

Bounds

- **Markov**:

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{2}{3}$$

- **Chebyshev**:

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{2}{n}$$

Conclusion

Chebyshev gives a **better n-dependent bound**.

12.6 Important Lecture Message

Tail bounds are used because:

- Exact tails are hard
- Bounds are easier
- Bounds still give guarantees
- They are crucial for **system reliability and overload analysis**

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13. Ultra-Short Exam Memory Sheet

If you need to revise in **2 minutes**, memorize this:

Tail

$$P(X > x)$$

Markov

$$P(X \geq a) \leq \frac{E[X]}{a} \quad (X \geq 0)$$

Chebyshev

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Binomial Tail

$$\sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson Tail

$$\sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Running Example

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right)$$

$$P\left(X \geq \frac{3}{4}n\right)$$

Markov:

$$\leq \frac{2}{3}$$

Chebyshev:

$$\leq \frac{2}{n}$$

Hierarchy

$$\mathrm{Markov} \to \mathrm{Chebyshev} \to \mathrm{Chernoff}$$

(Progressively tighter tail bounds.)

If you want, I can next convert this into either:

1. **clean LaTeX scribe format**, or
2. **very short viva/oral-exam revision notes**.

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concise but comprehensive, specially designed for effective exam revision.

Below are **formal, concise, and exam-focused lecture notes** for **Lectures 21 & 22: Inequalities and Tail Bounds**, strictly based on the provided slides/PDF. All definitions, formulas, results, and examples are included **without adding external material**. Source:

CSE400 – Lecture 21 & 22
Inequalities and Tail Bounds (Revision Notes)

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1. Tail of a Random Variable

Definition

The **tail** of a random variable X is:

$$P(X > x)$$

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2. Motivation for Tail Analysis

Applications

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean $\longrightarrow$ average behavior (easy)
- Tail $\longrightarrow$ rare extreme events (hard)

Difficulty

Tail computation requires summing **many small probabilities**.

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3. Exact Tail Computation

Binomial Tail

For $X \sim \text{Binomial}(n, p)$:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson Tail

For $X \sim \text{Poisson}(\lambda)$:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observation

Exact computation involves **long and complex sums**.

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4. Need for Tail Bounds

Idea

Instead of exact computation:

$$P(X \geq k) \leq \text{simple upper bound}$$

Advantages

- Easier to compute
- Provides guaranteed bounds
- Enables safe approximation

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5. Running Example

Experiment

Flip a fair coin n times.

Random Variable

$$X = \text{number of heads}, \quad X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

Mean

$$E[X] = \mu = \frac{n}{2}$$

Tail Event

$$P\left(X \geq \frac{3}{4}n\right)$$

---

6. Markov's Inequality

6.1 Key Idea

- Uses only the **mean**
- Large values are rare

6.2 Statement

For a non-negative random variable $X \geq 0$ and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

6.3 Proof (Idea)

$$E[X] = \sum_{x=0}^{\infty} x p_X(x) \geq \sum_{x=a}^{\infty} x p_X(x) \geq \sum_{x=a}^{\infty} a p_X(x) = aP(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

6.4 Application to Running Example

$$\mu = \frac{n}{2}, \quad a = \frac{3}{4}n$$

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{\frac{n}{2}}{\frac{3}{4}n} = \frac{2}{3}$$

Observation

Bound is **constant (independent of n)**.

6.5 Real-Life Example: Income

$$E[X] = 10{,}000$$

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000} = 0.2$$

6.6 Solved Example 1 (Exam Grades)

$$E[X] = 70, \quad a = 90$$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9} \approx 0.778$$

6.7 Solved Example 2 (Server Load)

$$E[X] = 5000, \quad a = 20{,}000$$

$$P(X \geq 20{,}000) \leq \frac{5000}{20{,}000} = \frac{1}{4} = 0.25$$

6.8 Limitation

Markov inequality applies only to **right tail**:

$$P(X \geq a)$$

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7. Chebyshev's Inequality

7.1 Key Idea

- Uses **mean + variance**
- Controls deviation from mean

7.2 Statement

For any random variable X with mean μ and variance $\mathrm{Var}(X)$:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

7.3 Proof Idea

Apply Markov to $(X-\mu)^2$:

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$
$$\leq \frac{E[(X-\mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

7.4 Running Example

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right)$$
$$\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$
$$P\left(X \geq \frac{3}{4}n\right) = \frac{1}{2}P\left(|X - \frac{n}{2}| \geq \frac{n}{4}\right)$$
$$\leq \frac{1}{2} \cdot \frac{\frac{n}{4}}{(\frac{n}{4})^2} = \frac{2}{n}$$

Insight

Bound **decreases with** n.

7.5 Real-Life Example (Exam Scores)

$$\mu = 70, \quad \sigma = 10$$
$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

7.6 Solved Example (Grades with Variance)

$$\mu = 70, \quad \sigma = 5, \quad \sigma^2 = 25$$
$$X \geq 90 \Rightarrow |X-70| \geq 20$$

$$P(X \geq 90) \leq \frac{25}{400} = \frac{1}{16} = 0.0625$$

Comparison

- Markov: 0.778
- Chebyshev: 0.0625

Chebyshev gives a **tighter bound**.

---

8. Chernoff Bound and Poisson Tail

Coverage in Lecture

- Chernoff Bound
- Poisson Tail
- Definition, proof, and example (as listed in lecture outline)

Poisson Tail Expression

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Key Role

Chernoff bounds provide **tighter tail bounds** than:

- Markov
- Chebyshev

Hierarchy of Bounds

$$\text{Markov} < \text{Chebyshev} < \text{Chernoff}$$

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9. Jensen's Inequality

Concepts Covered

- Convex functions
- Jensen's inequality
- Proof and example

Key Idea

Relates expectation and convex functions of random variables.

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10. Case Study: System-Level Understanding

Problem

Predict worst-case demand on servers.

Insight

Tail probabilities capture:

- overload events
- bursts in demand
- extreme system behavior

Conclusion

System design must consider **tail behavior**, not only averages.

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11. Key Results Summary

Tail

$$P(X > x)$$

Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a} \quad (X \geq 0)$$

Chebyshev's Inequality

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Binomial Tail

$$\sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson Tail

$$\sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Running Example Bounds

$$P\left(X \geq \frac{3}{4}n\right)$$

- Markov:

$$\leq \frac{2}{3}$$

- Chebyshev:

$$\leq \frac{2}{n}$$

Final Insight

- Exact tails → hard
- Bounds → simple + guaranteed
- Essential for analyzing **rare extreme events**

This version is **complete, structured, and directly usable for exams**.
If you want, I can convert this into a **LaTeX scribe** or **1-page cheat sheet**.

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Absolutely — below is a **clean, structured, exam-focused lecture note set for Lectures 21 and 22**, written **strictly from the provided lecture PDF/slides only**, with **no outside additions**. It is designed so you can **revise directly for exam preparation**. Source:

CSE400 – Fundamentals of Probability in Computing
Lecture 21 & 22 Revision Notes
Inequalities and Tail Bounds

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1. Introduction to Tails

Definition: Tail of a Random Variable

The **tail** of a random variable X is:

$$P(X > x)$$

It represents the probability that X takes a **large value**, i.e., lies in the **extreme right region** of the distribution.

Why Do We Care About Tails?

Tail probabilities are important because they correspond to **rare but critical events** such as:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- **Mean** $\rightarrow$ captures average behavior (easy to compute)
- **Tail** $\rightarrow$ captures rare extreme events (hard to compute)

Why Are Tails Hard to Compute?

Because they often require summing **many very small probabilities**.

---

2. Tail Examples

Example 1: Random Job Assignment

Suppose n jobs are distributed among n servers uniformly at random.

Question

What is the probability that a particular server gets at least k jobs?

Model

$$X \sim \text{Binomial}(n, p)$$

Tail Probability

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Interpretation

This is a **tail event**, representing a rare overload on one server.

Example 2: Poisson Arrivals

Jobs arrive to a datacenter according to a Poisson process with rate λ jobs/hour.

Question

What is the probability that at least k jobs arrive during the first hour?

Model

$$X \sim \text{Poisson}(\lambda)$$

Tail Probability

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Interpretation

This is a **tail event** corresponding to a **burst of arrivals**.

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3. Why Tail Bounds Are Needed

Exact tail probabilities are often **difficult to compute** because they involve:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

or

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

These are **long and complicated sums**.

Main Idea of Tail Bounds

Instead of computing the exact tail probability, we compute an **upper bound**:

$$P(X \geq k) \leq \text{something simple}$$

Advantages

- Easier to compute
- Gives a valid guarantee
- Useful for safe approximation

---

4. Running Example Used in the Lecture

The lecture develops progressively better tail bounds using the same example.

Experiment

Flip a fair coin n times.

Random Variable

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right)$$

Mean

$$E[X] = \mu = \frac{n}{2}$$

Question

$$P\left(X \geq \frac{3}{4}n\right)$$

This is the probability of getting **unusually many heads**, i.e., a **tail event**.

---

5. Markov's Inequality

5.1 Key Idea

Markov's inequality uses only the **mean** of a non-negative random variable.

Interpretation

If the average is small, then **very large values must be rare**.

5.2 Statement

For a **non-negative random variable** $X \geq 0$ and any $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

This provides an **upper bound on the right tail**.

5.3 Proof

Start with the expectation:

$$E[X] = \sum_{x=0}^{\infty} x\, p_X(x)$$

Restrict to the region $x \geq a$:

$$E[X] \geq \sum_{x=a}^{\infty} x\, p_X(x)$$

Since $x \geq a$ for all such terms,

$$E[X] \geq \sum_{x=a}^{\infty} a\, p_X(x)$$

$$E[X] \geq a \sum_{x=a}^{\infty} p_X(x)$$

$$E[X] \geq a\, P(X \geq a)$$

Hence,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

5.4 Markov on Running Example

Given:

$$X \sim \text{Binomial}\left(n, \frac{1}{2}\right), \qquad E[X] = \frac{n}{2}$$

We want:

$$P\left(X \geq \frac{3}{4}n\right)$$

Applying Markov's inequality:

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{\frac{n}{2}}{\frac{3}{4}n}$$
$$= \frac{2}{3}$$

Observation

This is a **very loose bound**, because it does **not depend on** n.

5.5 Real-Life Example: Income Distribution

Scenario

Average income:

$$E[X] = 10{,}000$$

Question

What fraction of people earn at least $50{,}000$?

Using Markov:

$$P(X \geq 50{,}000) \leq \frac{10{,}000}{50{,}000}$$
$$= 0.2$$

Result

At most **20%** of people can earn $50{,}000$ or more.

5.6 Solved Example 1: Exam Grades

Problem

Average exam grade is 70%.
"A" grade cutoff is 90%.

Find an upper bound on the fraction of students who receive an “A”.

Solution

Let $X \geq 0$ denote the student grade.

$$E[X] = 70$$

Using Markov’s inequality:

$$P(X \geq 90) \leq \frac{70}{90}$$

$$= \frac{7}{9} \approx 0.778$$

Result

At most **77.8%** of students can receive an “A”.

5.7 Solved Example 2: Server Request Load

Problem

A server processes an average of $5{,}000$ requests per second.

Find an upper bound on the probability that it receives $20{,}000$ or more requests in a given second.

Solution

Let:

$$X = \text{requests/second}$$

Then:

$$E[X] = 5000, \qquad a = 20{,}000$$

Using Markov:

$$P(X \geq 20{,}000) \leq \frac{5000}{20{,}000}$$

$$= \frac{1}{4} = 0.25$$

Result

At most **25%** chance of receiving $20{,}000$ or more requests.

5.8 Homework Problem (Markov)

Problem

Average annual income in a city is $40{,}000$.

(a)

Use Markov's inequality to find an upper bound on:

$$P(X \geq 120{,}000)$$

(b)

Can Markov's inequality be used to bound:

$$P(X < 10{,}000)$$

Lecture Hint

Markov bounds only the **right tail**:

$$P(X \geq a)$$

It does **not directly bound the left tail**.

---

6. Chebyshev's Inequality

6.1 Key Idea

Chebyshev's inequality uses:

- **Mean**
- **Variance**

It bounds the probability that a random variable deviates **far from its mean**.

6.2 Statement

For a random variable X with mean μ and variance $\mathrm{Var}(X)$, for any $a > 0$,

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Equivalent notation:

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

where $\sigma^2 = \mathrm{Var}(X)$.

6.3 Proof Idea

Apply Markov's inequality to the non-negative random variable:

$$(X-\mu)^2$$

Then:

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Applying Markov:

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

Since:

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

we get:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

6.4 Chebyshev on Running Example

For:

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right)$$

we have:

$$\mu = \frac{n}{2}, \qquad \mathrm{Var}(X) = \frac{n}{4}$$

We want:

$$P\left(X \geq \frac{3}{4}n\right)$$

From the lecture:

$$P\left(X \geq \frac{3}{4}n\right) = \frac{1}{2}P\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right)$$

Applying Chebyshev:

$$P\left(\left|X - \frac{n}{2}\right| \geq \frac{n}{4}\right) \leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Thus:

$$\begin{aligned} P\left(X \geq \frac{3}{4}n\right) &\leq \frac{1}{2} \cdot \frac{4}{n} \\ &= \frac{2}{n} \end{aligned}$$

Key Insight

This bound **decreases as n increases**, so it is **better than Markov**.

6.5 Real-Life Example: Exam Scores

Scenario

Mean exam score:

$$\mu = 70$$

Standard deviation:

$$\sigma = 10$$

Question

What fraction of students score below 40 or above 100?

This is:

$$P(|X - 70| \geq 30)$$

Applying Chebyshev:

$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2}$$
$$= \frac{100}{900} = \frac{1}{9}$$

Result

At most $\frac{1}{9}$ of students are this far from the mean.

6.6 Solved Example: Exam Grades with Variance

Problem

Average grade is 70%, standard deviation is 5%, and "A" cutoff is 90%.

Apply Chebyshev's inequality to obtain a tighter upper bound on the fraction of "A" students.

Solution

Given:

$$\mu = 70, \qquad \sigma = 5, \qquad \sigma^2 = 25$$

Observe:

$$X \geq 90 \Rightarrow |X-70| \geq 20$$

Thus:

$$P(X \geq 90) \leq P(|X-70| \geq 20)$$

Applying Chebyshev:

$$P(|X-70| \geq 20) \leq \frac{25}{20^2}$$
$$= \frac{25}{400} = \frac{1}{16} = 0.0625$$

Result

$$P(X \geq 90) \leq 0.0625$$

This is **much tighter** than Markov's bound 0.778.

---

7. Chernoff Bound and Poisson Tail

The lecture outline explicitly includes:

- **Chernoff Bound and Poisson Tail**
- Definition and proof
- Example

Poisson Tail

For:

$$X \sim \text{Poisson}(\lambda)$$

the upper tail is:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

This corresponds to the probability of seeing **many arrivals** in a fixed time interval.

Progression of Tail Bounds

The lecture develops **progressively tighter tail bounds**:

$$\text{Markov} \to \text{Chebyshev} \to \text{Chernoff}$$

Thus, Chernoff is presented as a **stronger / tighter tail bound**.

---

8. Jensen's Inequality

The lecture outline explicitly covers:

- **Jensen's Inequality**
- **Convex Functions**
- Proof and Example

Convex Function

Jensen's inequality is stated in the context of **convex functions**.

Role in the Lecture

Jensen's inequality is included as an important **inequality tool** relating:

- random variables,
- expectation,
- and convex functions.

---

9. Case Study and System-Level Understanding

The lecture concludes with a **case study** focused on:

Question

How to predict the worst-case demands on server?

System Interpretation

Tail probabilities are used to estimate rare but important events such as:

- server overload,
- sudden spikes in traffic,
- extreme queue buildup,
- rare performance failures.

Main Engineering Insight

Average behavior alone is **not sufficient** for system design.
One must also understand:

$$P(X \geq \text{large threshold})$$

because system failures are often caused by **tail events**.

---

10. Final Exam Summary

Important Definitions

Tail

$$P(X > x)$$

Right Tail

$$P(X \geq a)$$

Deviation from Mean

$$P(|X - \mu| \geq a)$$

Exact Tail Forms

Binomial

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Poisson

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Markov's Inequality

Condition

$$X \geq 0$$

Formula

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Uses

- Mean only
- Simple
- Loose bound

Chebyshev's Inequality

Formula

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Uses

- Mean + variance
- Controls spread around mean
- Better than Markov in many cases

Running Example

For:

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right)$$

$$E[X] = \frac{n}{2}, \qquad \mathrm{Var}(X) = \frac{n}{4}$$

Tail of Interest

$$P\left(X \geq \frac{3}{4}n\right)$$

Markov Bound

$$\leq \frac{2}{3}$$

Chebyshev Bound

$$\leq \frac{2}{n}$$

Overall Message of Lectures 21 & 22

- Exact tails are often hard to compute
- Tail bounds provide simpler guaranteed estimates
- These are crucial for studying **rare extreme events** in computing systems

---

11. Ultra-Short Revision Sheet

If revising quickly, remember:

$$P(X > x)$$

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

$$X \sim \mathrm{Binomial}\left(n, \frac{1}{2}\right), \quad E[X] = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$$

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{2}{3} \quad (\mathrm{Markov})$$

$$P\left(X \geq \frac{3}{4}n\right) \leq \frac{2}{n} \quad (\mathrm{Chebyshev})$$

$$\mathrm{Markov} \to \mathrm{Chebyshev} \to \mathrm{Chernoff}$$

If you want, I can now convert this into either:

- **clean LaTeX scribe format**, or
- **handwritten-style short exam notes**, or
- **viva-style oral preparation notes**.